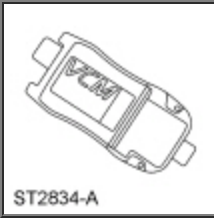


Module Configuration



Special Tool(s)	
	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

Configurable modules accommodate a variety of vehicle options, eliminating the need for many unique modules for one vehicle line. These modules must be configured when replaced as part of a repair procedure. Configurable modules should not be exchanged between vehicles since the settings are unique to each vehicle. Failure to configure a new module can result in improper operation and/or DTCs setting.

The following are the 3 different methods of configuration:

- Programmable Module Installation (PMI)
- Module reprogramming ("flashing")
- Programmable parameters

Some modules do not support all 3 methods.

Definition of Terms

The following are definitions of configuration terms:

Programmable Module Installation (PMI)

PMI is a scan tool process which configures settings in a new module. Data used for the PMI process is automatically downloaded from the original module and stored when a scan tool session is started. If this data cannot be retrieved from the module being replaced, the scan tool may prompt for As-Built data entry or display a list of parameter values that need to be manually selected. Some modules are reprogrammed during PMI when a strategy/calibration update is available. To carry out PMI , refer to Programmable Module Installation (PMI) in this section.

NOTE: It is important that the scan tool identifies the vehicle and obtains configuration data prior to removing any modules. The new module must be able to communicate with the scan tool in order to carry out PMI .

Module Reprogramming

Module reprogramming (also referred to as "flashing") is a scan tool process which updates the strategy/calibration in a module. Reprogramming a module with the same level of software does not improve module operation or repair a hardware failure. Module reprogramming is automatically carried out during PMI when a later strategy/calibration is available.

NOTE: Limit module reprogramming to circumstances where a published TSB procedure recommends doing so.

NOTE: A module cannot communicate with other modules on the communication network while being reprogrammed. After the reprogramming process, clear any network communication DTCs which may have been set in other modules.

NOTE: Some modules are reprogrammed in coordination with other modules. Follow the Integrated Diagnostic System (IDS) key cycling instructions carefully to avoid reprogramming errors, including failure of programming one or more of the coordinated modules.

Accessory Protocol Interface Module (APIM) Programming

APIM programming is a process that updates the APIM Consumer Interface Processor (CIP) and Vehicle Interface Processor (VIP) software. The VIP programming updates the calibration files in the portion of the APIM that interfaces with the Controller Area Network (CAN) . The CIP programming updates calibration files in the portion of the APIM that interfaces with mobile phones and other customer devices.

APIM programming uses the VCM to read and program the VIP software through the Data Link Connector (DLC) . A web-based On-Line Automotive Service Information System (OASIS) application reads and programs the CIP software via a Universal Serial Bus (USB) cable between the scan tool and vehicle USB port. Both the VCM and the Universal Serial Bus (USB) cable must remain connected, and OASIS application must be running on the scan tool during APIM programming. Carry out APIM programming. Refer to the appropriate section in Group 415 for the procedure.

Programmable Parameters

Programmable parameters are customer preference items that may be modified by the dealer via the scan tool, or in some cases, by the customer following a procedure listed in the vehicle Owner's Literature. While many configuration options may exist for a module, only a few of these options are programmable parameters.

Adaptive Learning and Calibration

Some modules require a separate learning procedure be carried out if replaced as part of a repair procedure. For adaptive learning and calibration instructions, refer to the specific module removal and installation procedures.

Vehicle Identification (VID) Block

Some PCMs contain a memory area called a Vehicle Identification (VID) block. The PCM VID block commonly stores powertrain configuration items such as Vehicle Identification Number (VIN) , tire size, axle ratio, and whether or not the vehicle is equipped with speed control.

Module Address

A unique module address is assigned to each module on the network for identification.

As-Built Data

As-Built data is a VIN -specific module configuration record. During vehicle build, the configuration from all modules is downloaded and stored in the As-Built database. As-Built data does not reflect customer preference items that have been changed from the default state. These items need to be changed using programmable parameters after the module is configured.

NOTE: It is not necessary to obtain As-Built data unless directed to do so by the scan tool. This data may be accessed from the technician service publication web site.

Module Configuration and Parameter Chart

Module	Requires PMI	Reprogram/Flash Capable	Requires Adaptive Learning or Calibration	Available Programmable Parameters
ABS module	Yes	Yes	Stability control initialization	Hill start assist enabled

Module	Requires PMI	Reprogram/Flash Capable	Requires Adaptive Learning or Calibration	Available Programmable Parameters
Accessory Protocol Interface Module (APIM)	Yes	Yes	No	Stereo in pass-through
Audio Front Control Module (ACM)	Yes	Yes	No	None
Audio Digital Signal Processing (DSP) module	Yes	Yes	No	None
Body Control Module (BCM)	Yes	Yes	No	- Axle ratio - Autolamps delay time - Front turn lamp outage configuration - Fog lamps present - Cruise control - Police dark car feature - Police rear door enable/disable - Smart unlocking - Rear turn lamp outage configuration - Tire placard pressure - Tire size - Daytime running lamps
Driver Seat Module (DSM)	Yes	Yes	No	None
Dual Climate Controlled Seat Module (DCSM)	Yes	Yes	No	None
Front Controls Interface Module (FCIM)	No	Yes	No	None
Front Control/Display Interface Module (FCDIM)	Yes	Yes	No	None
Front Display Interface Module (FDIM) (without navigation)	Yes	Yes	No	None
Global Positioning System Module (GPSM)	No	Yes	No	None
HVAC module — DATC	Yes	Yes	No	None
HVAC module — EMTC	Yes	Yes	No	None
Instrument Panel Cluster (IPC)	Yes	Yes	No	- Trailer Brake Control (TBC) gain - Trailer sway - Fuel tank selection (3.5L only) - Fuel capacity

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Module	Requires PMI	Reprogram/Flash Capable	Requires Adaptive Learning or Calibration	Available Programmable Parameters
OCSM	No	Yes	• Seat weight sensor re-zero	None
Parking Aid Module (PAM)	No	Yes	No	None
Power Steering Control Module (PSCM)	Yes	Yes	No	None
PCM	Yes	Yes	• Adaptive airflow • Idle speed • Refueling event • Fuel trim	• Speed control
Power Running Board (PRB) module	No	Yes	No	None
Restraints Control Module (RCM)	Yes	Yes	• Seat weight sensor re-zero • Stability control initialization	• Driver seat Belt-Minder® activation
Steering Column Control Module (SCCM)	Yes	Yes	No	None
Trailer Brake Control (TBC) module	No	No	No	None
Transfer Case Control Module (TCCM)	No	No	No	None
Tire Pressure Monitor (TPM) module (also known as Radio Transceiver Module [RTM])	No	Yes	No	None

Inspection and Verification

This section provides step-by-step module configuration procedures. Carry out the Programmable Module Installation (PMI) procedure in this section when another workshop manual section directs you to carry out configuration or when DTCs from the following chart are present:

DTC Chart

NOTE: *Some modules on this vehicle utilize a 5-character DTC followed by a 2-character failure-type code. The failure-type code provides information about specific fault conditions such as opens, or shorts to ground. Continuous Memory DTCs have an additional 2-character DTC status code suffix to assist in determining DTC history.*

DTC	Description	Source	Action
B2477	Module Configuration Failure	• ABS module • Dual Climate Controlled Seat Module (DCSM) • Driver Seat Module (DSM)	CARRY OUT Programmable Module Installation (PMI) . REFER to Programmable Module Installation (PMI) in this section.

DTC	Description	Source	Action
		Front Display Interface Module (FDIM)	
B2900	VIN Mismatch	ABS module	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
B6050	No Application Present	DCSM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
P0602	Powertrain Control Module Programming Error	PCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
P0605	Internal Control Module Read Only Memory (ROM) Error	PCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
P0630	VIN Not Programmed or Incompatible - ECM/PCM	PCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
P1639	Vehicle ID Block Corrupted, Not Programmed	PCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U0300	Internal Control Module Software Incompatibility	PCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U0300:00	Internal Control Module Software Incompatibility: No Sub Type Information	Restraints Control Module (RCM)	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2014:41	Control Module Hardware: General Checksum Failure	- Audio Front Control Module (ACM) - Audio Digital Signal Processing (DSP) module	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2014:41	Control Module Hardware: General Checksum Failure	Front Controls Interface Module (FCIM)	INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure.
U2014:42	Control Module Hardware: General Memory Failure	- ACM - Audio DSP module	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2014:42	Control Module Hardware: General Memory Failure	FCIM	INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure.
U201A:51	Control Module Main Calibration Data: Missing Calibration	- ACM - FCDIM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U201A:51	Control Module Main Calibration Data: Not Programmed	FCIM	INSTALL a new FCIM . REFER to Refer to the appropriate section in Group 415 for the procedure..

DTC	Description	Source	Action
U201A:51	Control Module Main Calibration Data: Not Programmed	• Audio DSP module	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2050	No Application Present	• ABS module • DSM • EDIM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2050	No Application Present	• PRB module	INSTALL a new PRB module. REFER to Section 501-08.
U2051	One or More Calibration Files Missing / Corrupt	• ABS module • EDIM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	• BCM • ACM • Audio DSP module • FCDIM • Instrument Panel Cluster (IPC) • HVAC module • Power Steering Control Module (PSCM) • RCM • Steering Column Control Module (SCCM)	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	• GPSM	INSTALL a new GPSM . Refer to the appropriate section in Group 415 for the procedure.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	• APIM	CARRY OUT APIM programming. Refer to the appropriate section in Group 415 for the procedure.
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	• ACM • Audio DSP module • FCDIM • Steering Column Control Module (SCCM)	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	• GPSM	INSTALL a new GPSM . Refer to the appropriate section in Group 415 for the procedure.
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	• APIM	CARRY OUT APIM programming. Refer to the appropriate section in Group 415 for the procedure.
U2200:00	Control Module Configuration Memory Corrupt: No Sub Type Information	• RCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2200:55	Control Module Configuration Memory Corrupt	• BCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.

DTC	Description	Source	Action
U2300:54	Central Configuration: Not Configured	Transfer Case Control Module (TCCM)	INSTALL a new Transfer Case Control Module (TCCM) . REFER to Section 308-07A.
U2300:55	Central Configuration: Not Configured	Transfer Case Control Module (TCCM)	INSTALL a new Transfer Case Control Module (TCCM) . REFER to Section 308-07A.
U2300:55	Central Configuration: Not Configured	- PSCM - RCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U2300:56	Central Configuration: Invalid/Incompatible Configuration	Transfer Case Control Module (TCCM)	INSTALL a new Transfer Case Control Module (TCCM) . REFER to Section 308-07A.
U3000:41	Control Module: Calibration / General Checksum Failure	- ACM - Audio DSP module - ECIM - HVAC module - PSCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U3000:41	Control Module: Calibration / General Checksum Failure	Parking Aid Module (PAM)	INSTALL a new PAM . REFER to Section 413-13.
U3000:41	Control Module: Calibration / General Checksum Failure	- ECIM - GPSM	INSTALL a new ECIM or GPSM . Refer to the appropriate section in Group 415 for the procedure.
U3000:41	Control Module: Calibration / General Checksum Failure	Transfer Case Control Module (TCCM)	INSTALL a new Transfer Case Control Module (TCCM) . REFER to Section 308-07A.
U3000:41	Control Module: Calibration / General Checksum Failure	APIM	CARRY OUT APIM programming. Refer to the appropriate section in Group 415 for the procedure.
U3000:46	Control Module: Calibration / Parameter Memory Failure	Transfer Case Control Module (TCCM)	INSTALL a new Transfer Case Control Module (TCCM) . REFER to Section 308-07A.
U3000:46	Control Module: Calibration / Parameter Memory Failure	- PSCM - RCM	CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.
U3000:46	Control Module: Calibration / Parameter Memory Failure	PAM	INSTALL a new PAM . REFER to Section 413-13.
U3000:54	Control Module: Missing Calibration	QCSM	INSTALL a new QCSM . REFER to Section 501-20B.
U3000:54	Control Module: Missing Calibration	PAM	INSTALL a new PAM. REFER to Section 413-13.
U3002:62	Vehicle Identification Number: Signal Compare Failure	- RCM - HVAC module	The PCM Vehicle Identification Number (VIN) is not correct. CARRY OUT PMI . REFER to Programmable Module Installation (PMI) in this section.